

SERENE BABU

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CAREER OBJECTIVE

Full Stack Developer with 3 years of experience building modern SPA and API-driven web applications using React, TypeScript, JavaScript, Python, Node.js, Express and relational databases such as MSSQL and PostgreSQL. Experienced in developing REST APIs, secure backend services and reusable frontend architectures, with hands-on experience in AWS cloud services and Docker. Strong focus on clean, maintainable code and delivering reliable solutions.

PROFESSIONAL EXPERIENCE

Full Stack Developer, *Caddayn, Remote*

Sept 2025 - Present

- Built ASP.NET Core (.NET 8) Web APIs using C#, Entity Framework Core, LINQ, and SQL Server for ERP-based employee management system.
- Designed SQL Server schemas, stored procedures, implemented JWT authentication and role-based authorization.
- Developed REST APIs with pagination, filtering, and centralized error-handling middleware.
- Integrated APIs with the frontend and reduced deployment time by 40% using CI/CD pipelines and Docker.

Full Stack Developer Volunteer, *Orcawise, Dublin, Ireland*

Dec 2024 – Sept 2025

- Developed a GenAI-powered chatbot platform using React (TypeScript) and Python Flask APIs.
- Built responsive interfaces with dynamic message rendering and asynchronous state handling.
- Integrated OpenAI LLMs (GPT-3.5, GPT-4, GPT-4o) for conversational and structured AI responses.
- Worked with LangChain-based RAG pipelines using embeddings and FAISS vector storage.
- Optimised request flow and improved regeneration response handling logic.

Front-End Developer, *Springpeople Software Pvt Ltd.*

Nov 2022 - Aug 2023

- Developed a responsive online course platform using HTML5, CSS/SCSS, JavaScript, and jQuery.
- Developed flexible frontend components capable of rendering dynamically managed content within a headless CMS-style architecture.
- Implemented client-side validation, optimized frontend performance for Core Web Vitals compliance.

Software Development Engineer, *Cloudegic Inc*

Apr 2022 - Nov 2022

- Built full-stack travel platform with React (TypeScript) frontend, Node.js/MongoDB backend featuring admin/user dashboards, bucket lists, and real-time notifications
- Developed REST endpoints using Express.js handling authentication, CRUD operations, social features (like/comment/share/bookmark) with MongoDB aggregation pipelines
- Unit tested the application using Jest and React Testing Library, achieving 92% code coverage.
- Managed complex asynchronous application state using state using hooks, Context API, Redux and Redux-Saga.

EDUCATION

- **Msc in Computing (Enterprise Software Systems) -2:1 Honours** | SETU, Waterford, Ireland | Sept 2023 – Sept 2024

Thesis – Usability and Accessibility of websites: A case study of applying Design Thinking and WCAG standards

- Developed a React Typescript web application applying WCAG 2.2 and UX best practices.
- Conducted usability research using personas, wireframes, empathy maps, and user journey analysis.
- Tested accessibility compliance using IBM Equal Access Toolkit and Accessibility Insights.

Modules:

Agile software Development – Kotlin, Vue.js, PostgreSQL, Unit tests –Junit, UI Tests –Selenium, Integration Tests | Enterprise Web Development – React, Typescript, Cloud-native deployment, Automated CI/CD workflows|Cloud Computing – AWS - S3, Lambda, EC2, SNS, SQS, DynamoDB, API Gateway, CloudFront, Cloudformation

- **BTech in Electronics and Communication Engineering | 2017 – 2021**

KEY COMPETENCIES

- React | NextJs | JavaScript | TypeScript | HTML5 | CSS | Tailwind | Bootstrap | Unit Testing | Integration Testing | UI Testing
- Kotlin | Maven | npm | SQL | MongoDB | Python Flask | C# | ASP.NET Core (.NET 8) | Entity Framework Core | LINQ | AWS
- Docker containerization | Atlassian Jira | Git & GitHub | CI/CD Workflows & Deployment | AI-Assisted Development: Claude
- Code Reviews | Debugging | Error handling | Problem Solving | SOLID Principles | Clean Architecture

CERTIFICATIONS

- Design Thinking Certificate by Interaction Design Foundation
- UI Design Pattern for Successful Software by Interaction Design Foundation

AWARDS

- Technical Talk Competition by IEEE – First Position
- Innovation & Entrepreneurship Challenge by IEDC (Innovation and Entrepreneurship Development Cell) SCET - First Place

PROJECTS

Spaceplorer - <https://github.com/serenebabu2022/Spaceplorer>

- Built a full-stack space-data dashboard using **React, TypeScript, Node.js, and Express**, integrating multiple NASA Open APIs to present astronomy imagery, near-Earth asteroid data, and solar activity.
- Developed interactive data visualisations, asteroid search, date filtering, hazardous-object filtering, and pagination, with responsive UI, loading/error states, and API error handling.
- Implemented unit testing with Jest and React Testing Library across key frontend components and backend functionality to improve reliability and maintainability.
- Currently engineering an AWS-based performance optimisation for API bottlenecks, using Python-based AWS Lambda functions for data retrieval/processing and Amazon ElastiCache to cache frequently requested NASA data, reducing repeated external API calls and improving response latency